Hackathon Problem Statements — Detailed Description

Problem Statement 1

Presence-Pulse: Detecting and Reducing Phubbing Behavior

Overview
Smartphones have become an inseparable part of daily life, shaping how people communicate, work, and socialize. While they enable constant connectivity, they also introduce subtle behavioral challenges in face-to-face_toggle interactions. One such behavior is *phubbing*, where individuals unintentionally ignore people physically present around them in favor of their phones. This often happens automatically due to habit loops triggered by notifications, boredom, stress, or the need for stimulation. Over time, repeated micro-distractions can weaken communication quality, reduce emotional connection, and create social friction in families, friendships, classrooms, and workplaces.

This problem focuses on designing a **context-aware behavioral awareness system** that helps users recognize and reduce phubbing behavior without restricting phone usage or invading privacy. The system should support awareness and behavior change through intelligent detection and gentle intervention.

Why This Problem Matters

- Phubbing is strongly linked to reduced relationship satisfaction and decreased perceived empathy during conversations.
- Frequent notification-driven phone checking fragments attention and reduces the quality of in-person communication.
- Most individuals are unaware of how often they disengage from conversations to check their phones.

- Existing solutions such as screen-time dashboards provide data but rarely influence real behavior.
- There is an opportunity to build technology that improves **digital well-being through behavioral awareness**, not restriction.

Core System Idea

The solution should function as a **digital presence coach** that understands when social interaction is likely happening and detects behavioral signals indicating distraction. Instead of blocking apps or enforcing limits, the system should provide **timely, minimal, and context-sensitive nudges** that encourage mindful phone usage. The system should learn individual usage baselines and adapt interventions to the user's habits over time.

What the Solution Should Do

- Detect contextual signals such as repeated screen unlocks, short-duration app switching bursts, and notification-triggered phone usage during likely social situations.
- Learn a personalized baseline of normal phone interaction patterns and identify deviations that suggest attention drift.
- Recognize recurring behavioral triggers such as increased phone checking during meals, meetings, or stressful moments.
- Provide subtle nudges that interrupt habit loops without being intrusive or judgmental.
- Generate periodic reflection summaries that help users understand patterns in their behavior.
- Optionally support shared presence goals for families, friends, or teams.

Technical Direction

- Android UsageStats API for device-interaction tracking
- Bluetooth proximity sensing to detect nearby devices

- On-device machine learning classification models
- Behavioral analytics pipelines
- Reinforcement nudging or habit-loop interruption logic

Problem Statement 2

AI-Powered Intelligent Triage & Decision Support System

Overview
Healthcare professionals must continuously interpret large volumes of unstructured information, including patient intake descriptions, consultation notes, voice conversations, and diagnostic reports. This information is often fragmented and time-sensitive, requiring rapid understanding and prioritization. The cognitive effort involved in synthesizing these inputs can lead to documentation burden, communication delays, and occasionally misprioritized cases.

This problem focuses on building an **AI-powered triage assistant** that converts unstructured healthcare information into structured summaries and prioritization signals. The goal is not to automate medical decisions but to improve **situational awareness and workflow efficiency** for healthcare providers.

Why This Problem Matters

- Healthcare professionals face increasing documentation workloads that reduce time spent with patients.
- Important symptoms or risk indicators may be buried within long, unstructured text.
- Prioritizing cases quickly is critical in emergency and high-volume environments.
- AI can help transform fragmented information into structured insights that support decision-making.
- Responsible AI systems can improve workflow efficiency without replacing human judgment.

Core System Idea
The system should act as a **clinical information assistant** that summarizes incoming data, extracts medically relevant entities, and highlights urgency indicators. By organizing information into structured formats, the system can

help professionals quickly understand patient context and prioritize attention where it is most needed.

What the Solution Should Do

- Convert text or audio inputs into concise structured summaries.
- Extract symptoms, timelines, risk indicators, and key events using NLP techniques.
- Classify urgency levels based on extracted information.
- Assist documentation by organizing extracted insights into readable formats.
- Provide a dashboard view showing case priority and workflow status.
- Use synthetic or public datasets to ensure responsible development.

Technical Direction

- LLM summarization pipelines
- Named Entity Recognition (NER) for clinical information extraction
- Text classification models for urgency prediction
- Speech-to-text pipelines for audio inputs
- Dashboard visualization systems
- Responsible AI evaluation practices

Problem Statement 3

The Silent Spiral: Personal Mental-State Awareness Companion

Overview

Mental health challenges often emerge gradually through small behavioral and emotional changes that go unnoticed for long periods. Many individuals struggle to identify these changes early due to limited emotional vocabulary, social stigma, or lack of safe reflection tools. By the time someone recognizes a problem, it may already feel overwhelming.

This problem focuses on building a **self-awareness companion** that helps users observe emotional and behavioral patterns over time. The system should support reflection and awareness rather than diagnosis, therapy, or treatment recommendations.

Why This Problem Matters

- Many people find it difficult to describe or interpret their emotional states.
- Cultural stigma often discourages open discussion about mental well-being.
- Behavioral patterns such as sleep changes, routine disruption, or mood shifts are difficult to notice without tracking.
- Early awareness can help individuals take small, constructive steps before problems escalate.
- Technology can provide a private, non-judgmental reflection space.

Core System Idea

The solution should feel like a **trusted reflection companion** that helps users better understand their emotions and routines. It should encourage gentle self-discovery through journaling, pattern detection, and prompts that help users articulate what they are feeling.

What the Solution Should Do

- Provide guided journaling tools for emotional reflection.
- Detect emotional patterns in text entries using NLP.
- Identify behavioral trends over time.
- Expand emotional vocabulary through prompts and suggestions.
- Encourage small, actionable reflection habits.
- Maintain a non-clinical and culturally sensitive experience.

Technical Direction

- NLP-based emotion detection
- Journaling analytics
- Pattern recognition models
- Prompt-generation systems

Problem Statement 4

Crowdsourced Treatment Intelligence Platform

Overview
Patients often seek real-world treatment experiences online, looking beyond clinical descriptions to understand outcomes, side effects, costs, and recovery timelines. However, this information is scattered across multiple platforms such as forums, blogs, videos, and social media. Extracting useful insights requires time, effort, and the ability to distinguish anecdotal experiences from reliable evidence.

This problem focuses on building a **crowdsourced treatment intelligence platform** that aggregates public patient discussions and transforms them into structured, traceable insights.

Why This Problem Matters

- Patients want to learn from others' treatment journeys.
- Information about side effects and recovery timelines is often fragmented.
- Online health discussions mix personal experience with misinformation.
- Structured aggregation of treatment experiences can improve patient awareness.
- Transparent source traceability increases trust in insights.

Core System Idea
The system should function as an **experience intelligence engine** that collects public discussions, analyzes them using NLP techniques, and presents structured summaries of treatment experiences across different medical approaches.

What the Solution Should Do

- Aggregate public treatment discussions from multiple sources.

- Extract side effects, outcomes, and recovery timelines.
- Detect sentiment patterns across treatments.
- Identify combination therapies mentioned in discussions.
- Provide traceable references to original sources.
- Flag potential misinformation.

Technical Direction

- Web scraping pipelines
- NLP summarization
- Topic modeling
- Sentiment analysis
- Source credibility scoring

Problem Statement 5

Open Innovation

Overview
The open innovation category allows participants to build technology solutions for any meaningful real-world problem. The emphasis is on identifying a clear problem, designing a feasible solution, and demonstrating real impact through a working prototype.

Possible Domains

- Climate technology
- Education systems
- Accessibility tools
- Public safety
- Governance platforms
- Productivity tools
- Sustainability solutions
- Community applications

What Judges Will Look For

- Clear articulation of the problem being solved
- Evidence of real users or stakeholders
- Functional prototype demonstrating feasibility
- Measurable or visible impact
- Thoughtful system design and usability